

Fast Recovery with a Conventional Probe

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A relatively simple scheme is proposed for reducing, without loss of sensitivity, the ringdown time of a conventional, $50\ \Omega$ matched probe by about an order of magnitude. It is shown that the ringing properties following a pulse are grossly influenced by the impedance into which the probe operates, and that, while optimal damping may only be achieved under special circumstances, in general, considerable reductions may be obtained by selecting correctly the preamplifier input impedance and the length of coaxial cable between the probe and the preamplifier. Suggestions are made for damping the high-voltage portion of the ringdown, and a practical circuit (used in imaging studies at 5 MHz with a probe of quality factor 1000) is given by way of example.

INTRODUCTION

It has long been recognized that the utilization of the full bandwidth of a tuned NMR probe presents considerable difficulties when pulse techniques are employed. The act of exciting the probe in order to transmit causes the probe to ring following the rf pulse, and one must therefore wait until the ringing has decayed to a negligible value before attempting to collect data. Failure to do so results in spectra which are badly distorted, since the first few data points may be grossly in error. However, waiting for ringdown to cease results, perforce, in a reduction of available bandwidth. Degrading the quality factor of the probe by resistive damping reduces the ringdown, but, on the other hand, it usually degrades signal-to-noise ratio, too. Numerous attempts have been made (1–8) to overcome this problem, and while the present author has published elsewhere a solution to the difficulty (8), that solution has been justly criticized on the grounds that it is too complex for general use. An alternative approach is therefore given here, based on an idea presented in that article. However, a full understanding of the technique to be examined does require a knowledge of the rationale underlying probe design, and we therefore consider first why it is that a probe rings at all.

Consider the use of a coil (duly optimized in geometry and construction for the sample and magnet to hand) for production of the exciting B_1 field. The coil comprises inductance and resistance (some of which may be associated with the sample (9)) and, of course, must be connected to the transmitter in order to receive current alternating at the Larmor frequency. Now we wish such a connection to produce, for the specified coil, the maximum current (and hence B_1 field) for the power available from the transmitter. In other words, the probe coil must be “power matched” to the transmitter. As the transmitter usually has a source resistance of $Z_0 = 50\ \Omega$, this

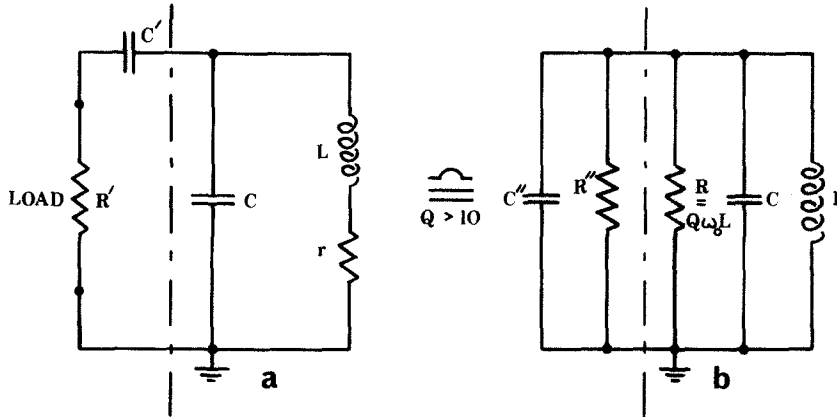


FIG. 1. One of several schemes which transform the predominantly reactive impedance of a probe coil to a resistance of $R' = 50 \Omega$ in order to effect power matching to the transmitter. The article explores what happens when R' is no longer 50Ω .

means that the impedance of the coil (for example, $j50 + 0.5 \Omega$, $j = \sqrt{-1}$) must somehow be transformed from a predominantly reactive (imaginary) value to a purely resistive value of 50Ω at the frequency of interest. There are several well-known ways of performing the transformation (10), but each involves the use of at least one capacitor, which, with its negative reactance, cancels the positive reactance of the coil, leaving purely resistance. Further manipulation is then required to make this resistance 50Ω , and a typical circuit is shown in Fig. 1a. If we plot with frequency the impedance Z of L , C , and r in the figure, and neglect, for the time being, the matching capacitance C' , we see from Fig. 2 that the curve is a familiar Lorentzian—a “resonance” centered at frequency

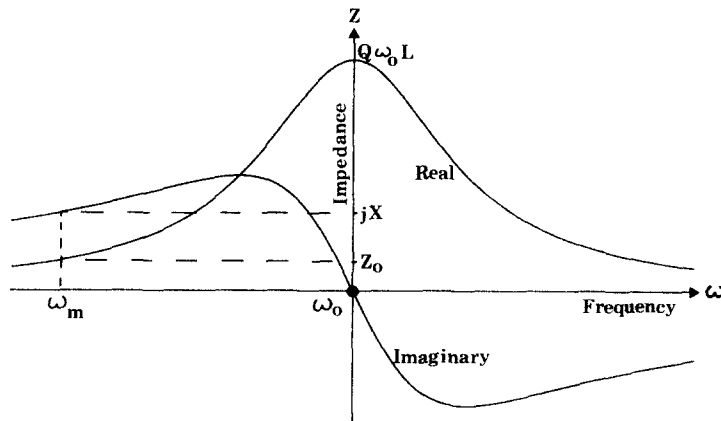


FIG. 2. The impedance of the tuned circuit, comprising L , C , and r in Fig. 1, plotted as a function of frequency. The curve is a Lorentzian, and therefore the circuit rings with an exponential decay pattern. At frequency ω_m , the real part of the impedance is 50Ω . The large positive reactance at this frequency is canceled by the addition of the matching capacitor C' , which provides an equal and opposite reactance. The remaining impedance is therefore purely 50Ω resistive.

$$\omega_0 = (LC)^{-1/2}. \quad [1]$$

Thus, if this circuit is subjected to a sudden electrical change, it rings, in the familiar manner of a FID, with a time constant which is related to the width at half height. In many cases the initial amplitude of the ringing can be greater than 1 kV in an NMR experiment, and before a signal may be received, this voltage must die down to the microvolt level. Now, as a separate matter, consider the plot of Fig. 2 at frequency ω_m , well below resonance. At this *particular* frequency, the real part of the Lorentzian is $50\ \Omega$, while the imaginary component is positive, and of value jX_m . If we now add the capacitor C' , whose reactance at this frequency is $-jX_m$ and calculable from the plot, the reactances cancel, and we are left with a resistance of $50\ \Omega$. Thus exact matching to $50\ \Omega$ occurs only at the one frequency ω_m with this simple circuit. On the other hand, the ringing of the circuit is dependent on the shape of the full frequency plot, and is related to it by its Fourier transform. The length of time Δt over which ringing persists is thus given by the uncertainty relationship

$$\Delta t \Delta \omega \geq 1 \quad [2]$$

where $\Delta \omega$ is the width of the plot at half height. The greater the Q factor of the circuit, the narrower the width and the longer ringdown lasts.

Now, when the receiver is connected to the system, the input impedance of the preamplifier (typically $500\ \Omega$ to $5\ \text{k}\Omega$) is between C' and ground. At frequency ω_m , this input impedance is very much larger than the $50\ \Omega$ of the probe and therefore has negligible effect on the noise performance of the system. However, as we depart from frequency ω_m , we see from Fig. 2 that the source impedance of the probe can change so much that the preamplifier input may be comparable or even relatively small, and hence no longer negligible. It follows that the impedance which the preamplifier presents to the probe, while negligible at frequency ω_m , can drastically alter the frequency response of the former away from ω_m , and thereby the ringdown characteristics of the system. This statement is the basis of the present study, and the rest of the article therefore deals with the investigation of this effect and its practical application. In short, signal-to-noise ratio and transmitting performance are analyzed and optimized at one frequency only; ringdown involves all frequencies, and can therefore be manipulated, without substantial degradation of sensitivity, by loading the probe, not with $50\ \Omega$, but rather with some optimal impedance.

DAMPING

Given that the probe coil and its associated capacitors (be they the circuit of Fig. 1a or some other configuration) are inviolate once they have been set to give $50\ \Omega$ at frequency ω_m , we must ask what impedance should be connected between the probe terminals (in place of R' in Fig. 1a) in order to minimize the ringing. This question is explored in the Appendix and unfortunately the optimal configuration, comprising a large inductance and a resistance, is rarely practical. However, connecting a simple resistance (R') of the right value can result in appreciable improvements. Using the series to parallel transformation indicated in Fig. 1b, the impedance that C' and R' present to the tuned circuit can be considered to be a resistance R'' in parallel with a capacitor C'' for

$$(R' + 1/j\omega C')^{-1} = \left(\frac{R'^2 \omega^2 C'^2 + 1}{R' \omega^2 C'^2} \right)^{-1} + \left(\frac{R'^2 \omega^2 C'^2 + 1}{j\omega C'} \right)^{-1}. \quad [3]$$

Hence

$$R'' = \frac{R'^2 \omega^2 C'^2 + 1}{R' \omega^2 C'^2}; \quad C'' = \frac{C'}{R'^2 \omega^2 C'^2 + 1}. \quad [4]$$

The tuned circuit, assuming that $Q \gg 1$, can likewise be transformed as shown, where the parallel resistance R is given by $Q\omega_0 L$. Note that the Q factor may now be expressed as the parallel resistance divided by $\omega_0 L$. To power match with $R' = Z_0 = 50 \Omega$, we set

$$R'' = Q\omega_0 L = R. \quad [5]$$

After a little algebra, and knowing that in practice, R'' must be $\gg 1$ (typically $> 10^3$), Eq. [5] reveals the well-known result (10) that

$$C' \simeq (C/Q\omega_0 Z_0)^{1/2} \quad [6]$$

where we have assumed that $R' \ll 1/\omega C'$ and $\omega_0 L = 1/\omega_0 C$. Thus, since Eq. [6] is correct, we have confirmation that C' is correctly adjusted when Eq. [5] is satisfied. In addition, we see that in the equivalent circuit of Fig. 1b, setting $R'' = R$ has halved the net parallel resistance, and thus halved both the effective Q of the circuit and the ringdown time. Now to obtain the most rapid ringdown, we would like to make R'' as small as possible. From Eq. [4] it may be seen that R'' is not a minimum when $R' = Z_0 = 50 \Omega$, but rather when $R' = 1/\omega C'$. In this situation, $R'' = 2/\omega C'$, and it is easy to show from Eq. [6] that for reasonable values of Q , Z_0 , and $1/\omega C$, $R'' \ll R$, e.g., by a factor of 5 or more. The effective Q value of the circuit is then

$$Q_{\text{eff}} \simeq R''/\omega_0 L = 2 \left(\frac{QZ_0}{\omega_0 L} \right)^{1/2}. \quad [7]$$

It follows that if the probe is terminated, not in 50Ω , but rather in a resistance of value $1/\omega C'$ (other than when the transmitter is operative) a substantial improvement in probe ringdown time (of the order of $\sqrt{Q}$) may be achieved. Further, from Eq. [6], because $1/\omega C'$ is much greater than 50Ω , the effect of this terminating resistance upon the noise performance is minimal. For example, a termination of 500Ω (which, in any case, may already be present in the preamplifier) degrades the noise figure of the spectrometer by 0.4 dB; a termination of $1 \text{ k}\Omega$ by 0.2 dB.

We must now incorporate the above result into the spectrometer design. If we consider for the moment only that portion of the ringdown during which the preamplifier is operative (i.e., from 1 V down), the probe is terminated by the input impedance of the preamplifier *as presented by the length of coaxial cable joining the two*. Clearly, the cable can completely change the effective impedance which loads the probe. For example, using 50Ω coaxial cable of length $\lambda/4$ between the preamplifier (of input impedance deliberately set to 500Ω) and the probe would result in the probe's termination by 5Ω —almost a short circuit. The effective quality factor of the probe would then be approximately its maximum value of $\omega_0 L/r$, and ringdown problems would be at their worst. Thus it would appear that the cable length joining the preamplifier and the probe should either be very short, or about $\lambda/2$ as shown in

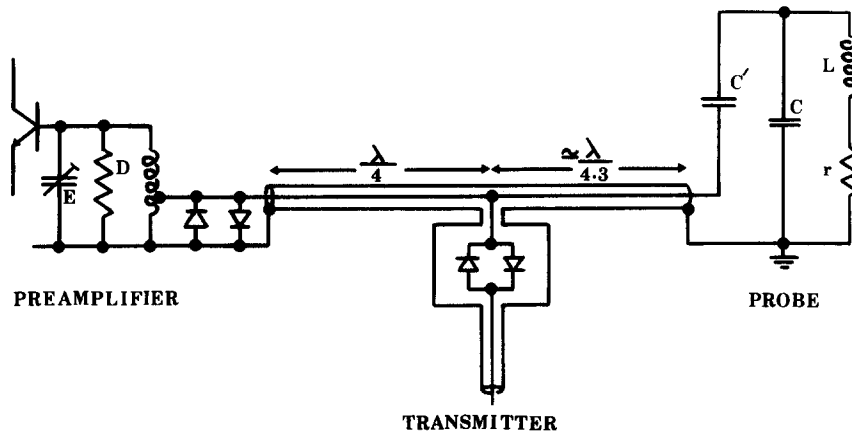


FIG. 3. A simple implementation of remote damping for the low-voltage ringing of the probe. The low Q ($\ll 10$) tuned circuit at the input of the preamplifier increases the $50\ \Omega$ resistance presented by the probe (via the coaxial cable) to that impedance which results in the best noise performance. However, at the probe, the preamplifier input impedance (as transformed by the cable) is that value which gives the best damping. Final trimming is performed by making changes in the resistor D , along with fine adjustment of capacitor E (such that the noise figure is not degraded) or adjustment of the length of cable. Note that this circuit does not improve the high-voltage ringdown.

Fig. 3. Figure 4 is an experimental plot of relative Q value, versus cable length and terminating resistance, for a probe at 60 MHz having the parameters already mentioned, namely $Q = 100$, reactance $50\ \Omega$, and cable characteristic impedance $50\ \Omega$. Equation [7] predicts a reduction in Q value of a factor of 5, and we may see from the figure that this value is slightly exceeded for an optimal resistance slightly less than $1/\omega C'$, and with a cable a little less than $\lambda/2$ in length. The reason for these variations is discussed in the Appendix.

HIGH-VOLTAGE DAMPING

So far, we have considered only that portion of the ringdown where the preamplifier is operative. However, the initial portion of the ringdown, which is at high voltage and from which the preamplifier is protected by, for example, crossed diodes (see Fig. 3), constitutes a serious problem. Unless the transmitter is of the tuned, class C variety, which is nowadays uncommon, it is likely that the transmitter loads the probe with $50\ \Omega$, and thereby only halves the effective Q value. A simple way of rapidly reducing the voltage is to apply a phase-inverted pulse of the appropriate length (δ) which cancels the ringing. However, employment of this trick requires accurate timing and probe tuning. An alternative is to use a PIN diode switch which replaces the transmitter with the optimal damping resistance, thereby rapidly quenching the ringing to the point where the preamplifier can take over. A suitable circuit designed for use with a 150 W transmitter at 5 MHz is shown in Fig. 5. The circuit can handle considerable potential, and during high-voltage ringing essentially places a low Q tuned circuit of the correct parallel resistance across the probe prior to switching in the preamplifier. The design has been used very successfully to reduce the ringdown

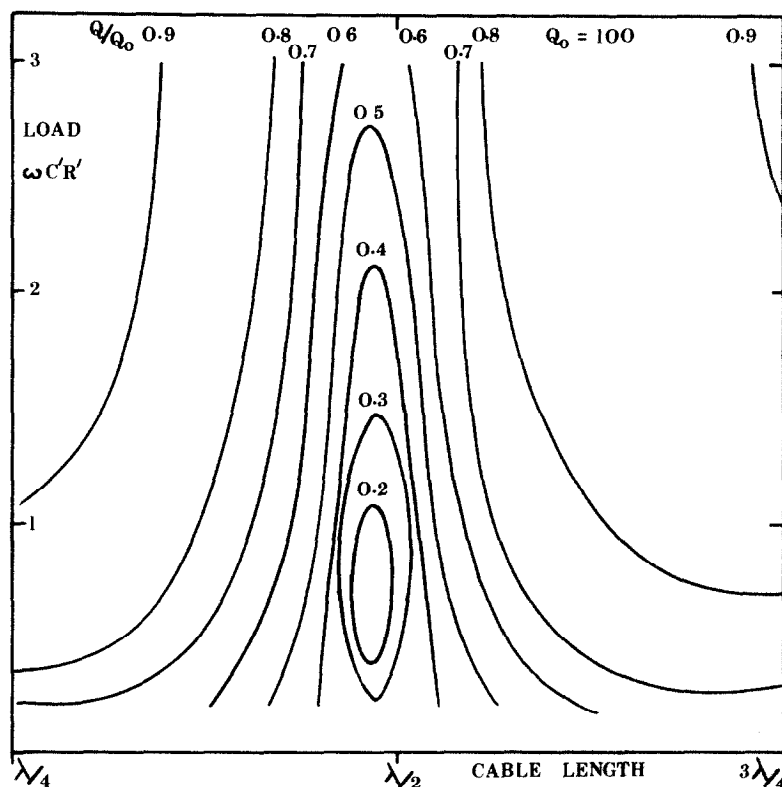
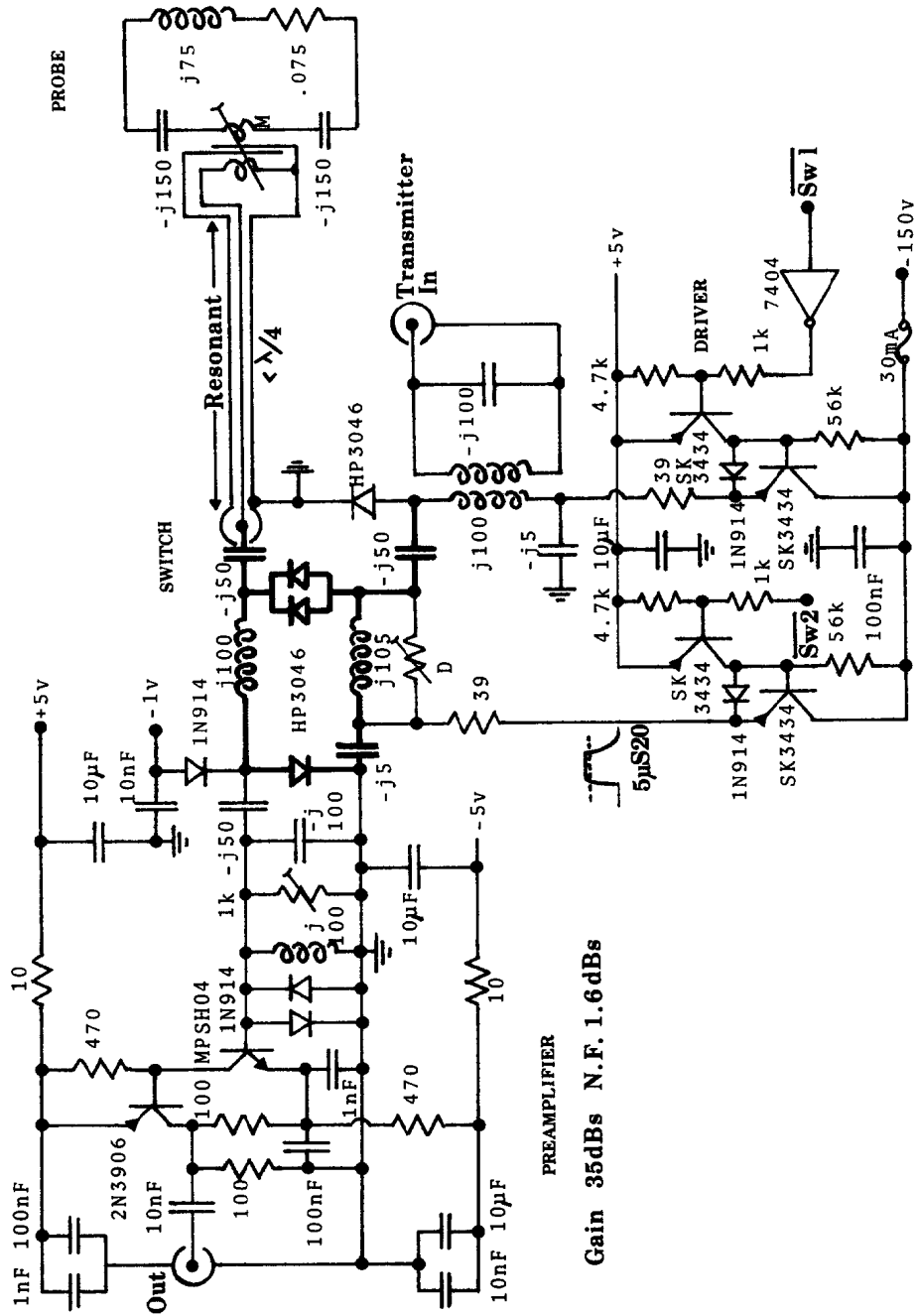


FIG. 4. A contour plot of the relative Q value when the probe of Fig. 1 is loaded with a resistance at the end of a $50\ \Omega$ coaxial line. The free-standing probe has a Q value of 100 and a reactance of $j50\ \Omega$, and the plot was performed at 60 MHz. By choosing the correct line length and terminating resistance, it is possible to reduce the effective Q value by at least a factor of 5.

time of a probe of Q value 1000 by a factor of 12. Note the use of transformer coupling which, in common with several other coupling methods, reduces the optimal line length to about $\lambda/4$.

An issue which has not yet been addressed is that of how much damping is actually needed. This depends, of course, very much on the details of the experiment, but one can make the general statement that as one departs from the matching frequency ω_m , so the effective probe impedance departs from $50\ \Omega$ and the preamplifier no longer delivers its best noise performance. How bad the degradation of performance may be is entirely dependent on the characteristics of the preamplifier, but it is normally fair to say that a bandwidth of no greater than $\Delta\omega = \omega_0/Q$ (the width of the absorption curve at half-height) may be employed. This bandwidth, in turn, dictates the fastest required ringdown. Turning the question around, we may ask what bandwidth is attainable, given the relationships between Q , bandwidth, and ringdown. The time constant of ringdown is related to the effective Q factor of the probe by the equation

$$\tau = 2Q_{\text{eff}}/\omega_0. \quad [8]$$



Thus from Eq. [7], when the probe is loaded for maximum damping,

$$\tau_{\min} \approx \frac{4}{\omega_0} \left(\frac{QZ_0}{\omega_0 L} \right)^{1/2}. \quad [9]$$

Now it may take as long as 20 time constants for the probe ringing to disappear into the background noise, and it is shown elsewhere (11) that the receiver may be "dead" for no more than 0.6 sampling periods if distortion of a spectrum is not to occur. It follows, from these numbers and Eq. [9] that

$$\Delta\omega_{\text{available}} \leq \frac{\omega_0}{20} \left(\frac{\omega_0 L}{QZ_0} \right)^{1/2} \quad [10]$$

and we may deduce that for typical values of Z_0 and $\omega_0 L$ of 50 Ω , the full bandwidth of the probe is only available if the Q factor is greater than 400. The damping scheme proposed is *proportionately* better the higher the Q value of the probe (as may be seen by taking the ratio of Eqs. [8] and [9]) which is just as well, as it is those probes with high Q which cause the greatest problems.

CONCLUSION

In conclusion, the thrust of the article lies in the realization that whereas both power (on transmission) and noise (on reception) matching are accomplished at *one* particular frequency, the ringdown time of a probe is dependent on its characteristics at *all* frequencies. The relationship between frequency and time behavior is characterized, as is usual with a Fourier conjugate pair, by the uncertainty relationship $\Delta\omega\Delta t \geq 1$. Thus it is possible, by tampering with the impedance which loads the probe (e.g., the preamplifier input impedance) to alter the effective bandwidth and hence reduce the ringdown time, while leaving substantially unaffected the noise performance at the one frequency at which the probe is matched to 50 Ω . For those able, the building of a PIN diode switch and a preamplifier with optimal input impedance can produce substantial improvements in ringdown time. However, just a few minutes spent adjusting the length of coaxial cable between the probe and the preamplifier may result in considerable gains. Finally, it is worth noting in Eq. [7] that it is best to use a receiving coil having the largest inductance possible, consistent with not compromising sensitivity by approaching self-resonance, and also the minimum transmitter source impedance—an unlikely variable!

FIG. 5. A full implementation of the ideas discussed in the text for use at 5 MHz with a high Q receiving coil. The probe is transformer matched, and only a short length of cable ($<\lambda/4$) is needed to resonate (with low Q factor) the matching coil. The PIN diode switch, which can handle 150 W, manipulates the various parts of a T section filter. The filter (heavy lines) has a characteristic impedance of 50 Ω and when operative upon transmission is essentially transparent. The PIN diodes transform the filter into a low Q tuned circuit of the desired parallel resistance when high-voltage damping is needed, and then into a series-tuned circuit which passes the probe signal to the preamplifier upon reception. The preamplifier has a noise figure of 1.6 dB and has an input impedance which is optimal for probe damping. The circuit decreases the overall recovery time by a factor of 12, with negligible loss of sensitivity.

APPENDIX

Optimal Damping of a 50 Ω Matched Probe

In the interests of practicality, we assume that whatever termination is applied to the matched probe of Fig. 1 results in a configuration having no more than four poles. Given this limitation, we may suspect that optimal damping may occur when capacitor C' is tuned out with an inductance L' , and the ensuing double-tuned circuit is then damped with a resistance r' in series with L' . Considering the current i flowing through L , some algebra shows that the differential equation governing the transient behavior of such a system is

$$\{D^4(LCL'C') + D^3(L'C'Cr + LCC'r') + D^2(LC + L'C' + LC' + CC'rr') + D(C'[r + r'] + Cr) + 1\}i = 0 \quad [A1]$$

where D is the differential operator d/dt . Let the resonant frequency of the basic probe be $\omega_0 = (LC)^{-1/2}$. Let

$$x = \omega_0^2 L'C', \quad y = \omega_0 C'r', \quad Q = \omega_0 L/r = 100, \quad \omega_0 L = 50 \Omega, \\ Z_0 = 50 \Omega \quad \text{and hence} \quad C' = C/10. \quad [A2]$$

In other words, we are taking the example used in the main text, namely a probe of Q value 100 and reactance of $j50 \Omega$. Try as a solution of Eq. [A1]

$$i = e^{k\omega_0 t}. \quad [A3]$$

We then obtain the auxiliary equation

$$k^4 x + k^3(y + 10^{-2}x) + k^2(1.1 + x + 10^{-2}y) + k(1.1 \times 10^{-2} + y) + 1 = 0. \quad [A4]$$

Searching numerically for the most rapid overall decay, i.e. for those values of x and y which maximize the magnitude of the real part of the smallest of the four roots, we find, not surprisingly, that the solution occurs when the four roots are equal, in which case

$$x \simeq 1, \quad y \simeq 0.63, \quad k \simeq -0.16 + j1.0. \quad [A5]$$

The general solution of Eq. [A1] is then of the form

$$i = e^{k\omega_0 t}(\alpha + \beta t + \gamma t^2 + \delta t^3) \quad [A6]$$

where α through δ are constants. Rather than evaluating Eq. [A6], we consider the practicality of the solution. As we suspected, as $x = 1$, L' is in resonance with C' . Thus, for our example, L' has a reactance of $j500 \Omega$. Without the use of ferrites, which is impractical near the NMR magnet, it is impossible to wind a pure inductance of this value—distributed capacitance is of major importance and the coil will be not too far removed from self-resonance. Even if such a coil could be made, only at the lowest frequencies would the first transistor of a preamplifier have a capacitance so small as to be negligible (i.e., reactance $\ll j500 \Omega$). Notwithstanding the above arguments, a bench experiment was attempted at 60 MHz using the values of Eq. [A5], viz., $\omega_0 L' = 500 \Omega$, $r' = 315 \Omega$. An impressive damping was obtained, corresponding to a Q of about 8 (R1 $k = 0.063$). However, placing one's fingers near L' increased the Q value substantially, showing that distributed capacitance was degrading

the performance. Variation of L' and r' confirmed that the values of Eq. [A5] were indeed optimal.

An interesting point arises concerning Fig. 4. The length of coaxial cable needed to terminate the probe in an inductance of $j500\ \Omega$ is 0.484λ , i.e., a little less than $\lambda/2$, as found experimentally in the figure. However, to present to the probe the optimal damping impedance of Eq. [A5], a cable of length 0.489λ terminated with $1.1\ \text{k}\Omega$ is required. The latter value is well removed from the experimental optimum. This is not surprising, for the cable is itself a resonant system and so a four-pole analysis is no longer valid.

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